Code No: 117CF **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, June/July - 2024
DESIGN PATTERNS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What is the primary purpose of Design Patterns in software development?	[2]
b)	What is the significance of organizing Design Patterns into a Catalog?	[3]
c)	Why is formatting a crucial aspect of document editing?	[2]
d)	What does Prototype pattern facilitate in terms of object creation?	[3]
e)	Define the Composite pattern.	[2]
f)	How does the Proxy pattern differ from the Flyweight pattern?	[3]
g)	Differentiate between the Chain of Responsibility and Command patterns.	[2]
h)	How does the Mediator pattern promote loose coupling between components?	[3]
i)	Define the Template Method pattern.	[2]
j)	What is the importance of discussing Behavioral Patterns in software design?	[3]

PART – B

(50 Marks)

2.a) Analyze how Design Patterns are integrated into the Smalltalk Model-View-Controller (MVC) architecture, highlighting their role in promoting modular and reusable code.
b) Evaluate the effectiveness of Design Patterns in addressing common design problems.[5+5]

OR

3.a) Illustrate how Design Patterns can be applied to improve the design and implementation of software systems in various domains.
b) Investigate the role of Design Patterns in promoting code reusability and maintainability. [5+5]

4.a) Evaluate the importance of document structure in a document editor.
b) Discuss the challenges associated with supporting multiple window systems in a document editor. [5+5]

OR

5.a) Discuss the Builder Pattern in detail providing an example.
b) Analyze the suitability of each Creational Pattern for designing a document editor.[5+5]

6.a) Compare and contrast the Decorator, Facade, Flyweight, and Proxy patterns.
b) Discuss the Adapter pattern in detail, including its intent, structure, participants, and scenarios where it is commonly used. [5+5]

OR

7.a) Discuss a real-world example where the Facade pattern can be applied.
b) Explore the Flyweight pattern, discussing its purpose of sharing common data across multiple objects to reduce memory consumption. [5+5]

8.a) Analyze the Command pattern in detail, outlining its components and how it decouples senders and receivers of requests.
b) Explain about the Chain of Responsibility pattern in detail. [5+5]

OR

9.a) Discuss the Mediator pattern, focusing on how it facilitates communication between objects without them being aware of each other's existence.
b) Evaluate the advantages and disadvantages of the Mediator and Observer patterns.[5+5]

10.a) Discuss the State pattern, including its structure, participants.
b) Discuss the Visitor pattern and its significance in separating algorithmic operations from the object structure on which they operate. [5+5]

OR

11.a) Explain about the Pattern Community an Invitation with an example.
b) Conclude with a parting thought on the significance of Design Patterns in software engineering. [5+5]

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